Low-cost LiDAR spatial reasoning: learned answers versus explicit geometric operations

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Status of the numbers in this version. Section 6 reports the *synthetic* campaign, under 45° octant sectors (`spatial_reasoning/results/all_results.json`). Work since has moved to real nuScenes-QA questions under that benchmark's six 60° sectors, a convention since checked against the gold answers rather than assumed. Under those conditions the measured figures differ from the ones below — the learned baselines do not reliably beat a scene-blind prior, and swapping ground-truth objects for detections barely moves them. The current state is tracked in `spatial_reasoning/results/state_of_results.md`, and the comparison this report argues for — one language model answering directly against the same model writing an executable program — is running at the time of writing and is not yet in these tables. The synthetic results are kept here because the argument they illustrate is unchanged; they should not be cited as results on nuScenes-QA.

Abstract

LiDAR-LLM [**?**] couples a frozen LiDAR encoder to a large language model so that spatial questions about a driving scene can be answered in natural language. Our original project reimplemented a reduced version of that architecture and showed, honestly, that it could not be trained on the hardware available: a GPT-2 answer head on filtered nuScenes-mini data, with no completed training run. This revised report changes the question rather than the hardware. Holding perception fixed — consuming 3D detections computed once and never retrained — we ask whether a spatial question is better answered by *learning* the answer or by *computing* it: parse the question into an explicit program, select the relevant objects, and apply a geometric operation. Four systems are compared under one protocol with scene-level splits: a question-only language prior, a deep-sets answer classifier, a differentiable counter, and a parameter-free symbolic program. On a balanced synthetic corpus the symbolic system reaches 0.916 accuracy against 0.656 for the best learned system — which does not beat the question-only prior — with a counting error of 0.13 against 0.53, no trained parameters, and three times less inference time. Contrast pairs make the gap starker still: 0.643 against 0.08 and below. Every symbolic error traces to perception, none to question understanding or geometry. The whole study runs on a CPU in under a minute per configuration, which is the point: the contribution is a comparison that is affordable, reproducible, and capable of a negative result.

1 Introduction

Interpreting a LiDAR scene in natural language is attractive for autonomous driving: a system that can answer "how many cars are ahead of us?" exposes its scene understanding in a form a human can audit. LiDAR-LLM [**?**] pursues this by aligning a LiDAR representation with the embedding space of a language model through a three-stage curriculum — cross-modal alignment, perception learning, and high-level instruction tuning — with GPT-4 [**?**] used to improve the captions that form the training data, and LLaMA-7B as the model that learns from LiDAR.

Our first report reimplemented a reduced version of this architecture. Section **??** restates what that implementation did and, more usefully, what it failed to establish. Section **??** then argues for a different question, one the same hardware can actually answer.

The shift is motivated by an observation about counting and directional questions. When the correct answer to "how many cars are in front?" is determined by the detected object list and a geometric predicate, asking a generative model to emit the digit is asking it to re-derive, inside its weights, a computation that is already exact and essentially free. Whether that re-derivation buys anything is testable, and this report tests it.

2 The original re-implementation and what it did not show

The first version of this project implemented the LiDAR-LLM pipeline in reduced form: a small PointNet as the LiDAR encoder, a view-aware transformer with six view-position embeddings, an MLP adapter projecting into the token embedding space, and GPT-2 [**?**] fine-tuned with LoRA [**?**] as the language model. Data came from nuScenes-mini [**?**] filtered to the samples present in the mini split, together with nuScenes-QA [**?**] questions: 323 training samples from 8 scenes and 81 validation samples from 2 scenes.

The honest summary is that it produced a debugged pipeline and no evidence. Training ran on laptops with roughly 6 GB of RAM and no GPU. The functional check was that the model could be driven to emit a fixed string regardless of the LiDAR input, which shows that gradients flow and nothing more. A distillation stage was implemented and its loss decreased, which again shows that the optimisation is wired correctly, not that the student retains any spatial ability.

Three weaknesses matter for what follows.

The evaluation could not separate competence from priors. nuScenes-QA answers are heavily skewed towards "no" and small integers. With no question-only control, an accuracy figure cannot distinguish a model that reads the scene from one that has learned the answer distribution. Real-3DQA [**?**] demonstrates how severe this is: on SQA3D a model fine-tuned on question-answer pairs *with no 3D input at all* matches or beats existing 3D-LLMs, and once the blind-answerable questions are filtered out, LEO's exact match collapses from 49.4% to 14.3% and Chat-Scene's to 17.0%.

The split invited leakage. Splitting questions rather than scenes places different questions about the same LiDAR sample on both sides of the split, so memorising "this scene has three cars" is rewarded.

The target task did not need a language model. Counting, existence and directional questions have answers that are functions of the object list. Nothing about them requires text generation.

3 A question the available hardware can answer

We fix perception and vary only the reasoning stage. Detections are computed once — CenterPoint-style 3D boxes [**?**], as shipped with nuScenes-QA, or ground-truth boxes for an oracle condition — and every system receives exactly the same object list. The research question is:

> *At fixed perception and comparable budget, is it better to learn the answer to a spatial question directly, or to select objects and then apply explicit geometric operations?*

This is narrower than "can LiDAR-LLM be made smaller?", and that is deliberate: it yields an interpretable result either way. If explicit computation wins, the learned head was spending

capacity on arithmetic. If the learned head wins, robustness to imperfect detections outweighs exactness — the more interesting finding, which the perturbation study of Section **??** is designed to detect.

Relation to existing work. Separating language understanding from geometric execution is not new, and our contribution is not the idea but its controlled measurement. ZSVG3D [**?**] already produced programs from language for zero-shot 3D grounding, composing view-independent, view-dependent and functional modules. KLDrive [**?**] is the closest system to the design tested here: it merges multi-source evidence into a scene knowledge graph and lets a frozen Qwen3-7B agent reason over it through a Plan–Execute–Observe loop restricted to a fixed action space, reaching 65.04% overall accuracy on nuScenes-QA and 64.46% on counting against 18.45% for its strongest baseline. Retrieve-to-Localize [**?**] makes the complementary argument for coordinates: rather than generating them as text, it retrieves language-conditioned spatial proposals and refines the position from local LiDAR points. BEV-LLM [**?**] extends LiDAR-LLM directly — the authors state that their method builds upon it — by adding camera images through BEVFusion and an absolute positional encoding, with a frozen Llama and a LoRA adapter. NuScenes-SpatialQA [**?**] builds spatial questions from nuScenes scene graphs and finds that a spatially enhanced VLM improves on qualitative questions while remaining uncompetitive on quantitative ones, which is precisely the counting gap this report targets. 3DGraphLLM [**?**] learns a scene-graph representation carrying semantic relations between objects, evaluated indoors on ScanRefer, Multi3DRefer, ScanQA, SQA3D and Scan2Cap.

Two things prevent a direct numerical comparison with our results. First, these systems change more than the reasoning module: KLDrive's accuracy is obtained with three detectors (RayDN, FocalFormer3D and IS-Fusion) over 39,704 selected questions under a random 8:2 split that is not stratified by scene, so part of the gain is perception and part is protocol. Second, "frozen LLM" does not mean cheap: KLDrive runs on RTX A6000 GPUs, Retrieve-to-Localize trains on four A6000s with 48 GB each, and BEV-LLM's 1B-parameter variant takes 48 hours on eight A100s despite freezing both BEVFusion and the language model. The experiment reported below runs in 49 seconds on a CPU. It is a much smaller claim, measured properly, rather than a larger claim taken on faith.

4 Method

4.1 Object-level scene representation

A scene is a list of objects, each with a canonical class (the ten nuScenes detection classes), a centre, a box size, a yaw and a detection score. All coordinates are in the ego frame: x forward, y left, z up. Detection results released for nuScenes are in global coordinates, so the conversion through the sample's ego pose is performed explicitly and in one place; a mistake there mirrors the scene and would be indistinguishable from a reasoning failure.

4.2 Directions as bearing sectors

Directional words are 45° bearing sectors centred on front, front-left, left, back-left, back, back-right, right and front-right. Sectors, rather than axis-aligned half-planes, give two properties we rely on: an object belongs to at most one named direction, and a car 40 m ahead and 2 m to the left is "front" rather than simultaneously "front" and "left". Objects closer than 1 m to the ego origin are excluded from directional queries, because a 10 cm localisation error flips the sign of their bearing.

Relative directions ("left of the truck") are ambiguous between the viewer's frame and the object's own frame. Both are implemented and the choice is an explicit parameter; the viewer-centric reading is the default, matching nuScenes-QA.

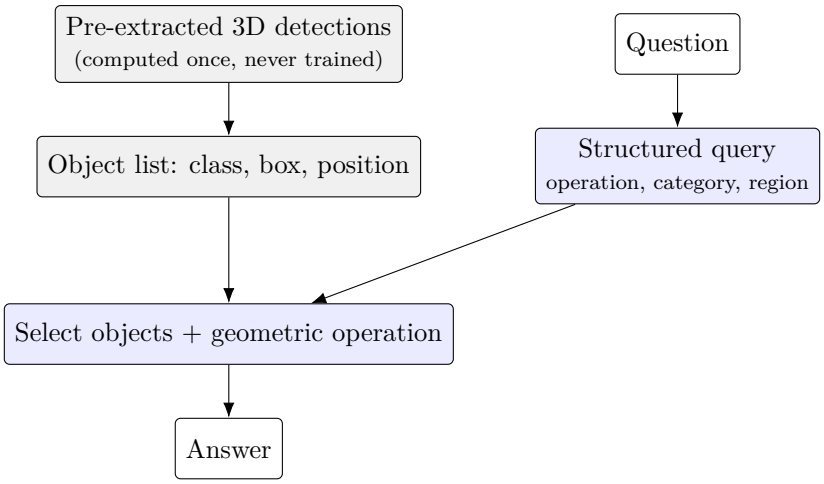


Figure 1: The symbolic pipeline. Grey is fixed perception, shared with every baseline; blue is the only part that varies between the compared systems. No language model appears anywhere.

4.3 From question to program

A rule-based parser maps a question onto a structured query: an operation (COUNT, EXISTS, COMPARE), a category or category group, and a region (anywhere, a direction sector, near, far, or a radius). The parser never sees the scene, which is what makes the error decomposition of Section **??** possible: a wrong answer is a comprehension error if and only if the parser produced the wrong query, independently of what the detector saw.

The parser *declines* questions outside its grammar instead of guessing. Coverage and accuracy are therefore separate numbers, and the results contain two symbolic variants: one that abstains, and one that falls back to the majority training answer so that it is directly comparable to a model which always emits something.

4.4 Execution

Executing a query is a filter followed by an arithmetic operation. It is deterministic, has no learned parameters, and returns the list of objects it selected, which is what allows an error to be attributed to perception rather than to reasoning. A definite reference whose resolution is ambiguous ("the truck", when the scene contains three) is refused rather than resolved by taking the first match: silently choosing would hide the failure and inflate the score.

4.5 The learned baselines

Three learned systems consume the same object lists.

Question-only.
A bag-of-words encoding of the question feeds an MLP over the answer vocabulary. It never sees the scene. Its accuracy is the language prior, and it is the threshold any other system must clearly clear before it can be said to use the 3D input.

Deep sets.
The question vector is concatenated to every object feature; a shared encoder is applied per object; sum and mean pooling are concatenated and classified. Sum pooling carries cardinality, which counting needs; mean pooling is scale-stable. This is the "predict the answer directly" condition — a miniature of the LiDAR-LLM answer head without the language model.

Deep sets + explicit counter.
The same backbone with an additional per-object relevance head in $[0, 1]$ whose sum is trained, with an L_1 loss, to equal the count. The inductive bias is "an answer of N comes from N objects", without hard-coding which ones.

The text encoder is deliberately the cheapest possible: CRC32-hashed unigrams and bigrams, no pretrained weights. The experiment is about where the computation happens, and a gain that required a large text encoder would confound that.

5 Experimental protocol

5.1 Data

Synthetic (used here). Scenes are sampled with nuScenes-like class frequencies and sizes, and questions are generated *from* queries, so the gold program of every question is known — which is what makes the comprehension/geometry split possible. A detector-failure model (12% drops, 5% duplicates, 0.6 m localisation jitter) turns the ground-truth list into the "detected" list the systems consume, so an oracle and a realistic perception condition exist for every sample. Ten scenes, 20 samples each, 8 questions per sample: 1600 questions over 200 samples.

A second, *degenerate* setting is reported alongside. Without balancing, 75% of answers mean "nothing is there" — a question about a rare class inside a 45° sector is almost always answered "none". The default setting caps that share at 35%. The comparison between the two settings is itself a result, and it reproduces Real-3DQA's finding on our own data.

nuScenes. The same code runs on released CenterPoint detections plus a nuScenes-QA style question file, with ground-truth boxes optionally supplied for the oracle condition. Those runs are not reported here.

5.2 Splits, metrics, probes

Splits are by scene, never by question. Accuracy is exact match, with an abstention counted as an error; for counting questions we also report mean absolute error, which separates "4 instead of 3" from "11 instead of 3". Cost is wall-clock time per item, measured identically for all systems; the one-off cost of running the detector is excluded from all systems equally. Three probes are run.

Contrast pairs. The same question text paired with two samples having different correct answers; a pair counts only if both are right. A system answering from the language prior cannot score here.

Counterfactual moves. One object is moved from one sector to another and the reference answer is recomputed. We report accuracy and how often the answer changed at all, because failing to react is the more diagnostic failure. The reference answers come from the same executor the symbolic system uses, so that system scores 1.000 by construction; the probe is evidence about the learned baselines and a regression test for the symbolic one.

Perception degradation. Detections are dropped, duplicated and jittered, one failure mode at a time. This perturbs detections, not the point cloud, and therefore models detector failures rather than sensor noise.

5.3 Error decomposition

Each symbolic error is attributed to a stage: **perception** (the same parsed program on ground-truth objects gives the right answer), **comprehension** (the parsed query differs from the gold query), **geometry** (query and objects both right, region maths wrong), **declined** (outside the grammar), or **unattributed** (no oracle boxes or gold program available).

6 Results

All figures are means over seeds $0, 1, 2$ with population standard deviations, test split $n = 320$ per seed. The balanced setting has 37.1% trivial answers and a majority-answer accuracy of 0.296; the degenerate setting has 75.0% and 0.412. Parser coverage is 1.000 in both, by construction: the questions were generated from the same grammar, so coverage here measures the instrument, not the language.

Table 1: Main comparison at fixed perception, balanced setting. All systems see the same detections and the same scene-level split.

System	Accuracy	Count MAE	Params	ms/item
Majority answer	0.296	—	0	—
Question only	0.656 ± 0.012	0.722 ± 0.093	35 863	0.68
Deep sets	0.655 ± 0.020	0.654 ± 0.088	120 855	0.88
Deep sets + counter	0.644 ± 0.018	0.534 ± 0.035	129 176	1.09
Symbolic	$\mathbf{0.916 \pm 0.027}$	$\mathbf{0.132 \pm 0.053}$	**0**	**0.23**

Three things stand out in Table **??**. The symbolic system is 26 accuracy points above the best learned system and its counting error is four times smaller, with no trained parameters and three times less inference time. Neither learned model beats the question-only control: deep sets matches it to within noise and the counter is marginally below it. And the counter does what it was designed to do, but only partially — it cuts the counting error from 0.654 to 0.534 without improving accuracy, so the inductive bias improves the magnitude of the estimate without making it exact.

Table 2: Consistency probes, balanced setting. Pair accuracy requires both members of a contrast pair to be correct. The symbolic system's counterfactual score is 1.000 by construction (Section **??**).

System	Pair accuracy	Counterfactual acc.	Answer changed
Question only	0.000	0.433	0.531
Deep sets	0.022	0.443	0.558
Deep sets + counter	0.076	0.406	0.521
Symbolic	0.643	$1.000^{\dagger}$	$1.000^{\dagger}$

Table **??** is the sharper result. A question-only model scores exactly zero on contrast pairs, as it must: given one question text and two different correct answers it can produce only one of them. Deep sets reaches 0.022 and the counter 0.076 — both far closer to the blind model than to the symbolic system's 0.643. On the accuracy table one could argue the learned models were using the scene a little; on contrast pairs they are essentially not using it at all.

The hypothesis that a learned counter would prove more robust than exact arithmetic is not supported. The symbolic system does degrade, and steeply under dropped detections — $0.916 \to 0.750$ at a 30% drop rate, by far the most damaging failure mode — while localisation jitter costs it almost nothing, as 45° sectors are coarse relative to a 3 m error. But it stays above every learned system at every level tested. The learned curves are flat, and that flatness is not

Table 3: Accuracy under detector degradation, balanced setting.

Setting	Question only	Deep sets	+ counter	Symbolic
clean	0.656	0.655	0.644	0.916
drop 10%	0.656	0.652	0.641	0.867
drop 20%	0.656	0.646	0.636	0.811
drop 30%	0.656	0.648	0.620	0.750
duplicate 10%	0.656	0.661	0.639	0.878
duplicate 20%	0.656	0.654	0.617	0.832
jitter 0.5 m	0.656	0.655	0.644	0.915
jitter 1.5 m	0.656	0.656	0.646	0.907
jitter 3.0 m	0.656	0.657	0.646	0.899

robustness: a model that ignores the object list cannot be harmed by corrupting it. Reading those two rows as "more robust" would be a mistake, and the contrast-pair column is what prevents it.

Table 4: Where the symbolic system's errors come from, balanced setting (mean counts over 320 test items).

Error kind	Count	Fraction
Correct	293.0	0.916
Perception	27.0	0.084
Comprehension	0.0	0.000
Geometry	0.0	0.000
Declined	0.0	0.000
Unattributed	0.0	0.000

Every symbolic error traces to perception: re-running the identical program on ground-truth boxes answers all of them correctly. Zero comprehension errors and zero geometry errors are the expected outcome on questions generated from the same grammar the parser implements, and should be read as a check that the instrument works rather than as a claim about natural language. On real nuScenes-QA text the comprehension row would be non-empty, and the value of the decomposition is that it would say by how much.

Benchmark design changes the conclusion. In the degenerate setting every number inflates and the differences compress: question-only reaches 0.757, deep sets 0.766, the counter 0.775 and the symbolic system 0.950. Reported alone, those figures would suggest the learned models do use the scene — they beat the blind model — when in fact they are riding a 75% base rate of "nothing there". Contrast-pair accuracy stays at 0.000, 0.045 and 0.063 respectively, exposing it. This is Real-3DQA's argument reproduced on our own data, and it is the strongest methodological lesson of the project: with a skewed answer distribution, a headline accuracy is close to uninformative.

7 Discussion

What the study establishes. On this task and this corpus, decomposing a spatial question into explicit operations is better on every axis measured: accuracy, counting error, consistency, parameter count and inference time. The margin is large enough that it does not depend on tuning. The learned baselines were given the same object lists and the same budget and did not clear a model that never looks at the scene.

What it does not establish. It does not establish an improvement to LiDAR-LLM. Demonstrating that requires integrating the module into the full system and evaluating against a comparable baseline; a result obtained with a small answer head does not transfer by assumption. The corpus is synthetic — no occlusion, no range-dependent class confusion, uniform layout — so it bounds what is worth trying on nuScenes rather than standing in for it. Parser coverage of 1.000 is an artefact of generating questions from the grammar; on real text it would drop, and the abstention column exists for that reason. And the restricted grammar means the symbolic system is not a general question answerer: outside counting, existence and comparison it has nothing to say.

If the architecture must stay LiDAR + LLM. The natural integration keeps a small frozen language model for the language interface and lets it call the executor, as KLDrive [**?**] does over its knowledge graph: the model decides *what* to compute, the executor decides *what the value is*. KLDrive's 64.46% counting accuracy against 18.45% for a direct-prediction baseline is consistent with what we measure at much smaller scale, and is the obvious next experiment.

Why the distillation line was dropped. The previous report proposed distilling the fine-tuned LLM into a smaller student. Distillation reduces the cost of a demonstrated capability; it cannot create one. Since no spatial competence was established in the teacher, distilling it would have measured compression quality against an unmeasured reference.

8 Conclusion

The original project established that the LiDAR-LLM architecture can be wired up on a laptop, and nothing more. Rather than repeat that attempt with a smaller language model, this revision asks whether, at fixed perception, decomposing a spatial question into explicit operations beats predicting the answer directly. On a balanced synthetic corpus it does, decisively and at zero training cost, and the learned alternatives fail to beat a blind baseline. The protocol was built to admit the opposite conclusion — a learned counter more robust to detector noise would have shown up in Table **??** — and it did not appear. The most transferable finding is methodological: the same systems compared on an unbalanced version of the same corpus look three times closer together than they are.

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